MILITARY SPECIFICATION SHEET

RELAYS, ELECTROMAGNETIC, ESTABLISHED RELIABILITY, DPDT,
LOW LEVEL TO 10 AMPERES, HERMETICALLY SEALED, ALL WELDED, DC COILS
(0.200 INCH TERMINAL SPACING, .062 DIAMETER TERMINAL PINS)
WITH INTERNAL DIODES FOR TRANSIENT SUPPRESSION AND POLARITY REVERSAL PROTECTION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-R-83536.

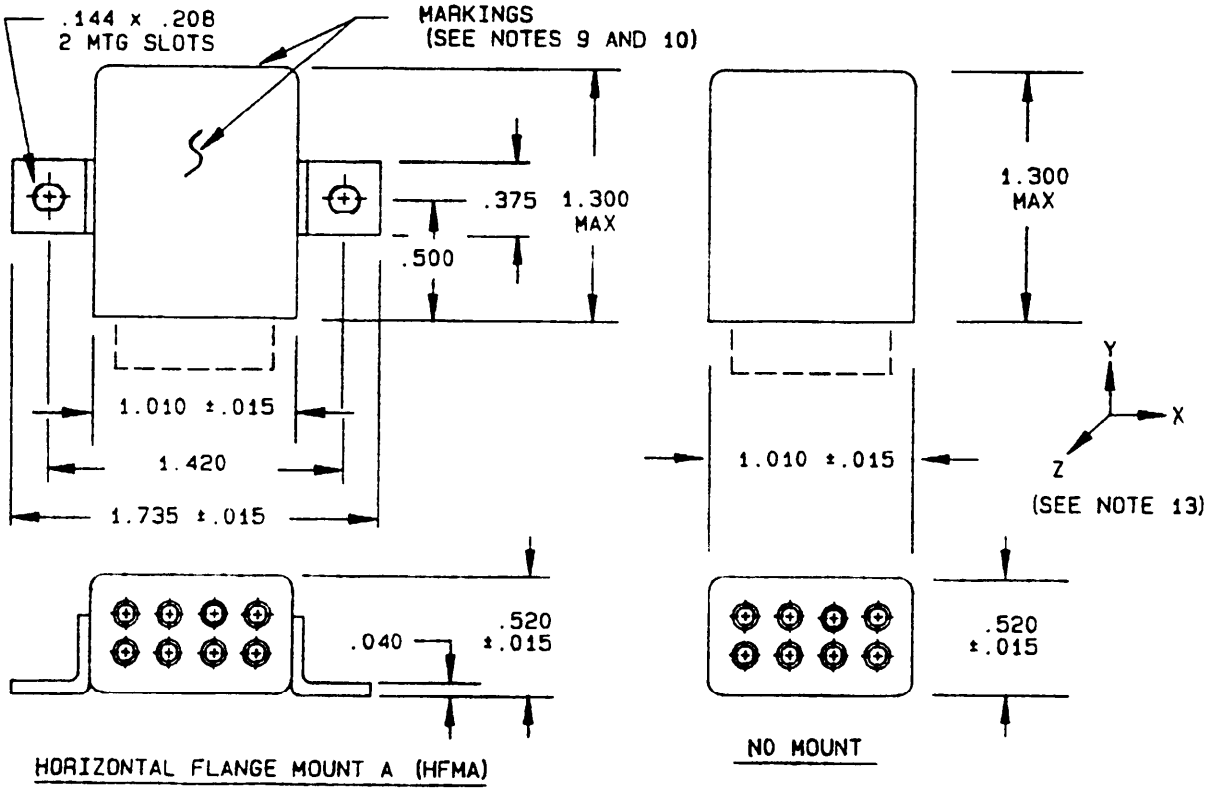
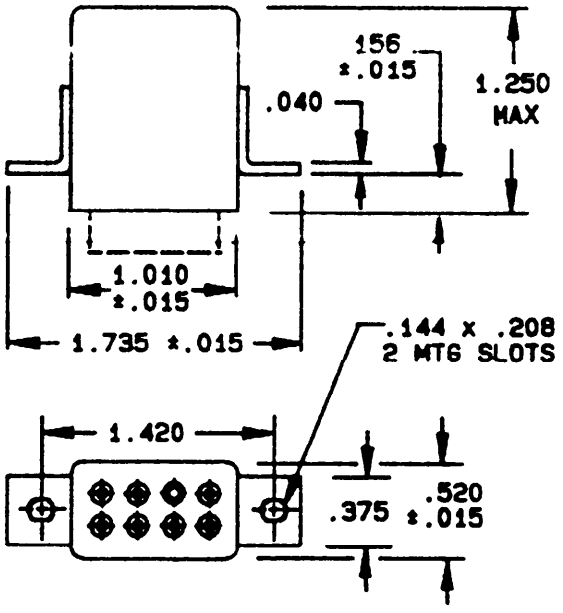


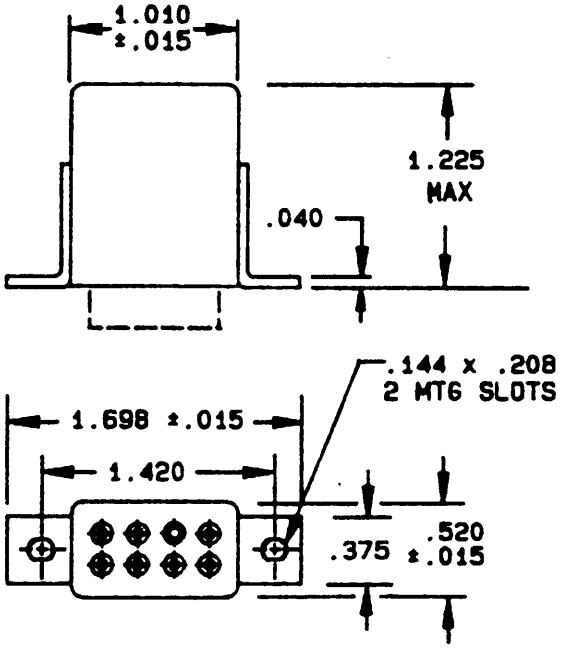
FIGURE 1. Dimensions and configurations.

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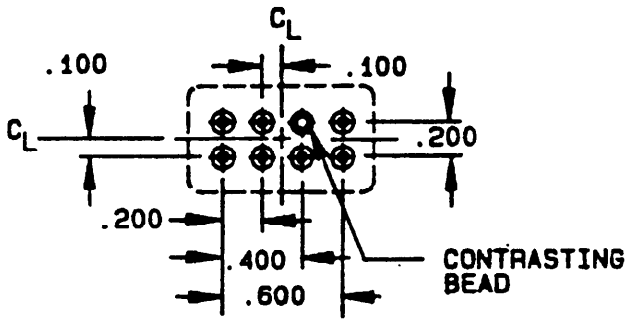
RAISED VERTICAL FLANGE MOUNT (RVFM)



VERTICAL FLANGE MOUNT (VFM)

FIGURE 1. Dimensions and configurations - Continued.

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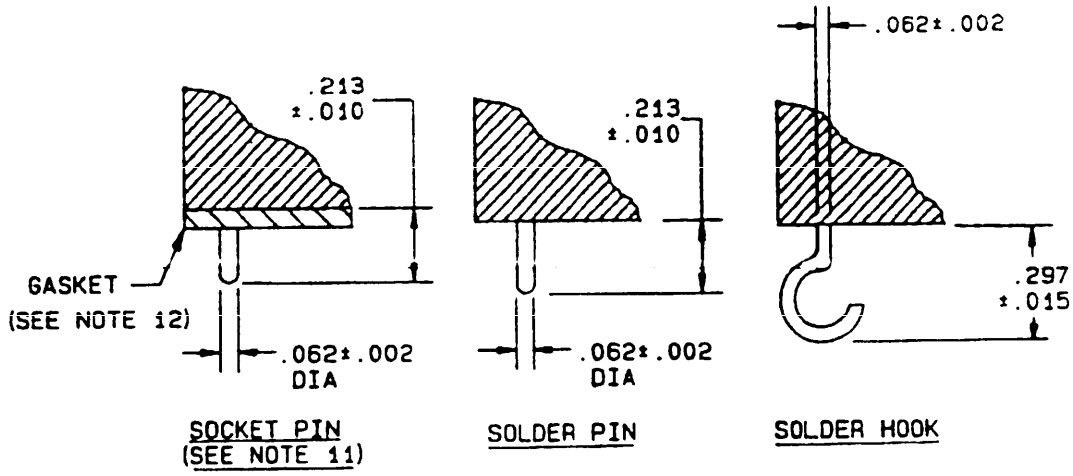


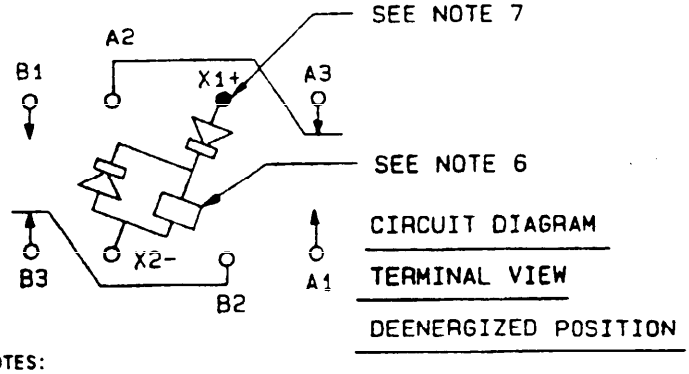
TERMINAL DIMENSIONS (TYPICAL)

FIGURE 1. Dimensions and configurations - Continued.

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Operate time: See table I.





Inches	mm	Inches	mm
.002	0.05	.213	5.41
.005	0.13	.297	7.54
.010	0.25	.375	9.52
.015	0.38	.400	10.16
.040	1.02	.500	12.70
.050	1.27	.520	13.21
.062	1.57	.600	15.24
.100	2.54	1.010	25.65
.144	3.66	1.420	36.07
.156	3.96	1.698	43.13
.200	5.08	1.735	44.07
.208	5.28		

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.010 (0.25 mm).
4. Terminal locating dimensions shown are applicable to all type mounts.
5. The shape and orientation of hook terminals is optional. Hook terminals to accommodate three number 22 AWG wires.
6. Coil symbol is optional in accordance with MIL-STD-1285.
7. Indicated terminal shall be identified with a contrasting bead.
8. There shall be affixed to the relay a suitable legible circuit diagram that identifies each terminal location specified.
9. Applicable to horizontal flange mount only. The circuit diagram, manufacturer's PIN, and the military PIN shall be marked on the near side. The remaining portion of the nameplate data shall be marked on the far side.
10. Applicable to horizontal flange mount only. Relays shall be marked with the manufacturer's name or CAGE code and date code. Marking shall be with the bottom of the print adjacent to the near side.
11. Socket pin terminals, except the polarizing pin, shall be gold-plated in accordance with MIL-G-45204, type II, class I. The nickel underplating shall be 50 to 150 microinches thick minimum.
12. Silicone rubber gasket, AMS 3332, shore hardness 15 to 35, thickness .050 ±.005 (1.27 ±0.13 mm).
13. Plane of critical motion for vibration and shock is Z-axis.

FIGURE 1. Dimensions and configurations - Continued.

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REQUIREMENTS:

Contact data:

Load ratings: See table I.

Low level: 10 to 50 μA at 10 to 50 mV dc or peak ac.

Intermediate current: Applicable.

Mixed loads: Applicable.

TABLE I. Rated contact load and cycles (amperes per pole).

Type of load (high level)	Cycles x 10^3	28 V dc	115 V ac 1 phase 400 Hz	115/200 V ac 3 phase 400 Hz	115 V ac 60 Hz 1/
Resistive	100	10	10	10	3
Inductive	100	4	4	4	2
Motor 2/	100	3	3	3	1.5
Lamp	100	2	2	2	0.5

1/ Life cycle rating 50,000 cycles.
2/ The relay shall be subjected to the specified life cycles for making five times the rated motor load at rated voltage, and breaking the normal rated load. Duration of the inrush current shall be 0.07 ±0.02 second, after which it shall be reduced to its rated motor load for the remainder of the on-time. Cycling rate shall be 0.35 ±0.09 second on and 2.0 ±0.1 seconds off.

Life: 100,000 cycles, unless otherwise specified (see table I).

Contact voltage drop or resistance:

Initial contact voltage drop: 0.100 volt maximum. Rated resistive current at 6 V dc or peak ac. Relays shall not make or break this load.

Initial contact resistance: 0.010 ohm maximum. 50 mA maximum at 6 V dc or peak ac.

High level life (contact voltage drop):

During life: 10 percent open circuit voltage maximum.

After life: 0.200 volt maximum.

Intermediate current (contact resistance):

During intermediate current: 3 ohms maximum.

After intermediate current: 1 ohm maximum.

Low level (contact resistance):

During low level: 100 ohms maximum.

After low level: 0.150 ohm maximum at 100 mA and 28 V dc.

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Contact bounce:

N/O: 2.0 ms maximum.

N/C: 5.0 ms maximum.

Contact stabilization time.

N/O: 2.5 ms maximum.

N/C: 5.5 ms maximum.

Overload current:

Resistive: 25 amperes at 28 V dc and 115 V ac, 400 Hz.

Inductive: Twice rated current, ac or dc

Coil data 1/: See table II.

Operate time: 15 milliseconds maximum with rated coil voltage at 25°C. 15 milliseconds maximum with rated coil voltage over the temperature range.

Release time: 15 milliseconds maximum from rated coil voltage at 25°C. 15 milliseconds maximum from rated coil voltage over the temperature range.

Duty rating: Continuous.

Diode characteristics:

Maximum negative transient: 1.0 volt.

Peak inverse voltage: 100 volts maximum.

Coil transient suppression: Applicable

Neutral screen: Not applicable.

Semiconductor in-process screening: Applicable. Visible inspection of semiconductors shall be in accordance with MIL-STD-750, method 2073 or 2074.

Electrical data:

Insulation resistance:

Initial: 100 megohms minimum at 500 V dc.

After life or environmental tests: 50 megohms at 500 V dc.

1/ CAUTION: Due to possible interaction of relay magnetic fields, the following spacing requirements, as a minimum, shall be considered in dense packaging situations:

a. Row to row assisting fields: .125 (3.18 mm).

b. Row to row opposing fields: .1875 (4.763 mm).

c. Side to side alternating fields: .0625 (1.588 mm).

d. Side to side like fields: .125 (3.18 mm).

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TABLE II. Dash numbers and characteristics. 1/

Dash number 2/ M83536/31-			Mount	Coil voltage (V dc) 3/		At 25°C				Over temperature range		
Socket pin	Solder pin	Solder hook		Rated	Max	Coil resistance (ohms) minimum	Specified pickup value (voltage) (V dc) 4/	Specified hold value (voltage) (V dc) 4/	Specified dropout value (voltage) (V dc) 4/	Specified pickup value (voltage) (V dc) 4/	Specified hold value (voltage) (V dc) 4/	Specified dropout value (voltage) (V dc) 4/
---	001	002	HFM	6.0	9.0	14.0	3.7	1.79	.40	5.0	2.5	.25
003	004	005	RVFM									
006	007	008	VFM									
009	010	---	No mount									
---	011	012	HFM	12.0	16.0	58	7.4	3.6	.80	10.0	5.0	.50
013	014	015	RVFM									
016	017	018	VFM									
019	020	---	No mount									
---	021	022	HFM	18.0	24.0	153	11.0	5.4	1.2	15.0	7.5	.75
023	024	025	RVFM									
026	027	028	VFM									
029	030	---	No mount									
---	031	032	HFM	28.0	32.0	270	14.7	7.1	1.8	20.0	10.0	1.0
033	034	035	RVFM									
036	037	038	VFM									
039	040	---	No mount									

See footnotes at end of table.

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TABLE II. Dash numbers and characteristics - Continued. 1/

Dash number 2/ M83536/31-			Mount	Coil voltage (V dc) 3/		At 25°C				Over temperature range		
Socket pin	Solder pin	Solder hook		Rated	Max	Coil resistance (ohms) minimum	Specified pickup value (voltage) (V dc) 4/	Specified hold value (voltage) (V dc) 4/	Specified dropout value (voltage) (V dc) 4/	Specified pickup value (voltage) (V dc) 4/	Specified hold value (voltage) (V dc) 4/	Specified dropout value (voltage) (V dc) 4/
---	041	042	HFM	48.0	52.0	900	29.5	14.3	3.6	40.0	20.0	2.0
043	044	045	RVFM									
046	047	048	VFM									
049	050	---	No mount									
---	051	052	HFM	120	122	6840	74.0	35.8	8.0	100.0	50.0	5.0
053	054	055	RVFM									
056	057	058	VFM									
059	060	---	No mount									

1/ Each relay possesses high level and low level capabilities. However, relays previously tested or used above 10 mA resistive at 6 V dc maximum or peak ac open circuits are not recommended for subsequent use in low level applications.
2/ The suffix letter L, M, P, or R to designate the applicable failure rate level shall be added to the applicable listed dash number. Failure rate level (percent per 10,000 cycles): L, 3.0; M, 1.0; P, 0.1; R, 0.01. Example: 001L, 002R.
3/ CAUTION: The use of any coil voltage less than the rated coil voltage will compromise the operation of the relay.
4/ Pickup, hold, and dropout voltages as shown are for test purposes only and are not to be used for design criteria.

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Dielectric withstanding voltage: 2/

	Sea level V rms (60 Hz)	Altitude V rms (60 Hz)
Between case, frame, or enclosure, and all contacts in the energized and deenergized position	1,000	500 All terminals to case
Between case, frame, or enclosure and coil	500	
Between all contacts and coil	1,000	
Between open contacts in the energized and deenergized positions	1,000	
Between contact poles	1,000	

Environmental characteristics:

Temperature range: -65°C to +125°C.

Shock (specified pulse): Applicable, MIL-STD-202, method 213, test condition C, except peak value shall be 100 g's for 6 ±1 ms. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Vibration (sinusoidal): Applicable, MIL-STD-202, method 204, 20 g's except frequency range shall be 10 to 2,000 Hz. Contact chatter shall not exceed 10 microseconds maximum for closed contacts and 1 microsecond maximum closure for open contacts.

Magnetic interference: Applicable.

Resistance to solvents: Applicable. Temperature of solvents A, B, C, and D shall be maintained between 63°C to 70°C.

Acceleration: 17 g's.

Physical data:

Dimensions and configurations: See figure 1.

Weight: .125 pound (56.7 g's) maximum.

Terminal strength:

Solder hook terminals:

Pull force: 3 pounds ±0.3 pound.

Bend: Applicable.

Twist: Not applicable.

Solder pin terminals:

Pull force: 3 pounds ±0.3 pound.

Bend: Applicable.

Twist: Not applicable.

2/ Dielectric may be improved by suitable insulation of terminals and wiring after installation.

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Socket pin terminals:

Pull force: 3 pounds ±0.3 pound.

Bend: Applicable.

Twist: Not applicable.

Terminal solderability: Applicable to solder pin and solder hook terminals only.

Seal: Hermetic.

Marking: Applicable.

Part or Identifying Number (PIN): M83536/31 (dash number from table II and suffix letter designating failure rate level). PIN is a new term encompassing previous terms used in specifications such as part number, type designator, and identification number.

Qualification inspection:

Qualification inspection and sample size: See table III.

TABLE III. Qualification inspection and sample size.

Single submission	Group submission	
38 units plus 1 open unit for level L at C = 0 1/	M83536/31-033	38 units plus 1 open unit for level L at C = 0 1/ Qualification inspection as applicable.
	M83536/31-030 M83536/31-031 M83536/31-032 M83536/31-038	2 units each part number, qualification inspection table, group II, shock, vibration (sinusoidal), acceleration, terminal strength, and seal.
	M83536/31-002 M83536/31-014 M83536/31-026 M83536/31-050 M83536/31-052	2 units, qualification inspection table, group II

1/ The number of units required for qualification testing will be increased as required in group V, table II of MIL-R-83536, if the relay manufacturer elects to test the number of units permitting one or more failures. Prior to performance of qualification testing, the relay manufacturer shall preselect the sampling plan.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Air Force - 99
DLA - ES

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5945-0824-29)

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